Autosomal Recessive Pseudohypoaldosteronism Type IB1 (PHA1B1, SCNN1A): Comprehensive Disease Characterization

Disease: Pseudohypoaldosteronism Type IB1, Autosomal Recessive (PHA1B1) **Causal gene: SCNN1A** (encoding the α-subunit of the epithelial sodium channel, ENaC) **Identifiers:** OMIM 264350 · MONDO:0009917 · Orphanet ORPHA:171876 (systemic PHA1) · ICD-10 E27.4 · MeSH D011546 (Pseudohypoaldosteronism) **Category:** Mendelian, autosomal recessive

Summary

Autosomal recessive **Pseudohypoaldosteronism type 1B1 (PHA1B1)** is a rare, neonatal-onset, **multisystem salt-wasting disorder** caused by biallelic loss-of-function mutations in **SCNN1A**, the gene on chromosome 12p13 that encodes the obligatory **α-subunit of the amiloride-sensitive epithelial sodium channel (ENaC)**. ENaC is a heterotrimeric (α:β:γ, 1:1:1) channel that serves as the terminal effector of aldosterone signaling in sodium-reabsorbing epithelia. When the α-subunit is lost, the channel cannot assemble or conduct Na^+, so target tissues become **resistant to aldosterone** despite grossly elevated hormone levels. The systemic (generalized) form contrasts with the milder, kidney-restricted autosomal dominant form (PHA1A) caused by heterozygous *NR3C2* (mineralocorticoid receptor) mutations.

Clinically, affected neonates present within the **first two weeks of life** with life-threatening **hyponatremia, hyperkalemia, and metabolic acidosis**, accompanied by **markedly elevated plasma renin and aldosterone** that are unresponsive to mineralocorticoid therapy. Because ENaC operates not only in the distal nephron and colon but also in **sweat glands, salivary glands, and the airway epithelium**, PHA1B1 produces extrarenal features that distinguish it from renal PHA1: **elevated sweat and saliva electrolytes** (overlapping with cystic fibrosis) and **cystic-fibrosis-like respiratory disease** driven by excess airway surface liquid and

impaired fluid absorption. The disorder is **lifelong**, strongly **associated with consanguinity**, and carries substantial neonatal mortality (~25% in one long cohort) and neurological morbidity from hyperkalemia-induced cardiac arrest.

There is no disease-modifying or curative therapy. Management is lifelong and supportive: **high-dose oral sodium supplementation**, **potassium removal** (cation-exchange resins/binders), and **correction of acidosis**, titrated to electrolytes. Severity tracks **residual channel activity** — truncating/null variants give the most severe, persistent disease, whereas hypomorphic missense or partially active variants can produce milder or even transient phenotypes that improve with age (partly through compensatory upregulation of the thiazide-sensitive Na-Cl cotransporter, NCC). This report compiles the genetics, mechanism, phenotype spectrum, diagnostics, prognosis, treatment, and model-organism evidence for PHA1B1, with primary-literature citations and ontology-term suggestions for knowledge-base curation.

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1. Disease Information

PHA1B1 is the autosomal recessive, **generalized (systemic / multisystem) form of pseudohypoaldosteronism type 1**, a condition of **end-organ resistance to aldosterone**. "Pseudo" denotes that the biochemical picture mimics aldosterone deficiency (salt wasting, hyperkalemia, acidosis) while aldosterone is in fact high — the defect lies downstream of the hormone, in the ENaC effector channel itself.

Key identifiers: - **OMIM:** 264350 (Pseudohypoaldosteronism type I, autosomal recessive) - **MONDO:** MONDO:0009917 - **Orphanet:** systemic/generalized PHA1 (ORPHA:171876); PHA1 parent ORPHA:79168 - **ICD-10:** E27.4 (other and unspecified adrenocortical insufficiency); **ICD-11:** mineralocorticoid-resistance endocrine disorders - **MeSH:** D011546 (Pseudohypoaldosteronism) - **Gene:** SCNN1A, HGNC:10599, NCBI Gene 6337, Ensembl ENSG00000111319, UniProt P37088 (SCNNA_HUMAN), locus 12p13.31

Synonyms: autosomal recessive PHA 1; PHA1B; autosomal recessive pseudohypoaldosteronism type 1; generalized/generalised PHA1; generalized/generalised pseudohypoaldosteronism type 1; multisystem/systemic PHA1.

Nomenclature note: "PHA1B" is used in OMIM for the recessive ENaC-subunit form as a whole. More granular subtyping assigns **PHA1B1 to SCNN1A**, **PHA1B2 to SCNN1B**, and **PHA1B3 to SCNN1G**. This report concerns the SCNN1A-driven subtype but necessarily draws on the shared ENaC biology, because most mechanistic and clinical literature treats the three subunit genes together.

Source of information: Derived almost entirely from **aggregated disease-level resources** (OMIM, Orphanet, reviews) and **published case reports / small single-center cohorts** rather than large EHR datasets, reflecting the disorder's rarity.

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2. Etiology

Primary cause — genetic. PHA1B1 is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function mutations in SCNN1A**. Of the three ENaC subunit genes, **SCNN1A-related cases are the most frequent** cause of systemic PHA1.

> PMID: 41457049: "The autosomal recessive systemic form (PHA1B), caused by mutations in SCNN1A, SCNN1B, or SCNN1G, is particularly severe and typically manifests in the neonatal period. Among these, SCNN1B-related cases occur less frequently than SCNN1A-related cases."

Genetic risk factors. - **Causal variants:** biallelic LoF variants in SCNN1A (frameshift, nonsense/premature-termination, splice-site; also hypomorphic missense). - **Consanguinity** is the dominant risk amplifier: it raises the chance of homozygosity for a recessive allele. A 37-year single-center cohort found **all affected children had consanguineous parents** (South Asian, Middle Eastern, or African origin) (PMID: 33690157). - **Founder effects** exist for specific alleles (e.g., a SCNN1G splice variant shared on a common haplotype across Indian-subcontinent families) (PMID: 8640238).

Environmental risk factors: None are causal. The disease is fully genetically determined. However, environmental/physiologic **stressors** (intercurrent illness, gastroenteritis, heat, salt restriction, dehydration) precipitate **acute salt-wasting crises** in already-affected individuals.

Protective factors. - **Genetic:** *residual channel activity* is protective in the sense of attenuating severity — missense/hypomorphic alleles that retain partial ENaC function yield milder or transient disease (PMID: 31301676; PMID: 20064610). Compensatory **NCC (SLC12A3)**

upregulation mitigates renal salt loss over time (PMID: 20376516). - **Environmental:** generous dietary **sodium** supplementation functionally rescues the salt-losing phenotype; this is therapeutic rather than preventive of the underlying disorder.

Gene–environment interaction: The principal interaction is between the **fixed genetic defect** and **sodium/potassium intake and physiologic stress** — a high-sodium, potassium-managing diet counteracts the channel defect, whereas salt restriction or illness unmasks crises.

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3. Phenotypes

PHA1B1 phenotypes fall into **laboratory abnormalities**, **clinical signs/symptoms**, and **extrarenal manifestations**. Onset is overwhelmingly **neonatal** (first days–weeks of life). Severity in systemic disease is **severe and persistent** (lifelong), contrasting with renal PHA1.

Phenotype	Type	Onset	Frequency	HPO term
Hyponatremia	Lab abnormality	Neonatal	~Universal	HP:0002902
Hyperkalemia	Lab abnormality	Neonatal	~Universal	HP:0002153
Metabolic acidosis	Lab abnormality	Neonatal	~Universal	HP:0001942
Elevated plasma renin	Lab abnormality	Neonatal	~Universal	HP:0000848 (hyperreninemia)
Elevated aldosterone (hyperaldosteronism)	Lab abnormality	Neonatal	~Universal	HP:0000859
Renal salt wasting	Clinical sign	Neonatal	~Universal	HP:0000127 (renal salt wasting)
Failure to thrive	Clinical sign	Neonatal/ infancy	Common	HP:0001508
Dehydration / volume depletion	Clinical sign	Neonatal (crises)	Common	HP:0001944
Vomiting / feeding difficulty	Symptom	Neonatal/ infancy	Common	HP:0002013
Recurrent respiratory infection / CF-like lung disease	Clinical sign	Infancy/ childhood	Common in systemic form	HP:0002783 / HP:0006532
Cough / wheeze / chest congestion	Symptom	Early childhood (≤5 y)	Frequent in young systemic patients	HP:0012735 / HP:0030828
Elevated sweat chloride / sweat electrolytes	Lab abnormality	Congenital	Characteristic of systemic form	HP:0012236
Skin lesions / eczematous dermatitis / miliaria	Physical manifestation	Infancy	Reported (salt deposition)	HP:0000964 (eczema)

Phenotype	Type	Onset	Frequency	HPO term
Cardiac arrhythmia / arrest (hyperkalemia)	Clinical sign	Neonatal crises	Minority but life-threatening	HP:0011675
Neurodevelopmental impairment (post-arrest)	Clinical sign	After crisis	~16% in one cohort	HP:0012759

Severity and progression: Systemic PHA1B1 is typically **severe and lifelong**, with **episodic** salt-wasting crises superimposed on a chronic course. Hypomorphic variants produce **milder, sometimes transient** disease that can improve with age.

Quality-of-life impact: Substantial. Affected children require lifelong oral salt and potassium-lowering medication, frequent electrolyte monitoring, and hospital admissions for crises; respiratory involvement adds CF-like morbidity. Survivors of hyperkalemic cardiac arrest may have permanent neurological disability. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data exist for this rare disorder.

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4. Genetic / Molecular Information

Causal gene. SCNN1A (sodium channel epithelial 1 subunit alpha), 12p13.31; HGNC:10599; OMIM gene 600228; disease OMIM 264350. The gene encodes the **α-ENaC subunit**, which is **obligatory** for channel assembly, trafficking, and Na^+ conduction. ENaC is a **heterotrimer of α (SCNN1A), β (SCNN1B, 16p12), and γ (SCNN1G, 16p12) subunits** in 1:1:1 stoichiometry.

> PMID: 38963175: "Systemic PHA type I (PHA1B) has varying degrees of clinical presentation and results from mutations in genes encoding subunits of the epithelial sodium channel (ENaC) including the alpha, beta, and gamma subunits. To date, about 45 variants of PHA1B have been identified."

Pathogenic variants. - **Variant types:** frameshift (including gross deletions, insertions/deletions), nonsense/premature termination, canonical splice-site, and missense. Truncating/null variants predominate in severe systemic disease; a **gross deletion of SCNN1A** has also been reported (PMID: 41275343). - **Classification (ACMG/AMP):** most reported SCNN1A LoF variants are **pathogenic/likely pathogenic**; novel missense variants are sometimes initially

VUS pending functional testing. - **Allele frequency:** causal alleles are **very rare** in gnomAD (private/family-specific or founder); LoF constraint on SCNN1A means null alleles are uncommon in the general population. - **Origin: germline**; no somatic involvement. - **Functional consequence: loss of function** — abolished or markedly reduced amiloride-sensitive Na^+ current. Truncating variants remove pore/gating elements; missense variants reduce open probability (e.g., a γ-ENaC A635V example reduced amiloride-sensitive current by ~30%, illustrating the residual-activity principle) (PMID: 40153144).

> PMID: 8589714: "Investigation of affected offspring of consanguineous union reveals mutations in either the alpha or beta subunits of the amiloride-sensitive epithelial sodium channel in five kindreds. These mutations are homozygous in affected subjects, co-segregate with the disease, and introduce frameshift, premature termination or missense mutations that result in loss of channel activity."

Modifier genes. NCC/SLC12A3 (thiazide-sensitive Na-Cl cotransporter) acts as a physiologic modifier — its upregulation partially compensates for ENaC loss. Broader aldosterone-pathway regulators (WNK kinases, Nedd4L/NEDD4L ubiquitin ligase, channel-activating proteases such as CAP1/Prss8, CAP3/St14, kallikrein-1) influence ENaC abundance/activity and are candidate modulators but are not established clinical modifiers in PHA1B1.

Epigenetic information: No disease-specific DNA-methylation or histone-modification findings are established for PHA1B1.

Chromosomal abnormalities: Point mutations and small indels dominate; **gross intragenic deletions** of SCNN1A occur but large cytogenetic rearrangements are not characteristic.

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5. Environmental Information

PHA1B1 is a **monogenic disorder with no environmental cause**. There are **no toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents** that cause the disease.

Environmental factors are relevant only as **modulators of the established disease:** - **Dietary sodium/potassium** — the key modifiable factor (therapeutic high-salt intake; potassium restriction). - **Heat/sweating** — excess sodium loss through defective sweat glands can worsen salt depletion. - **Intercurrent infection/gastroenteritis** — precipitates acute crises.

Infectious agents: none cause PHA1B1. Notably, recurrent lower-respiratory infections occur as a *consequence* of airway ENaC dysfunction, but — unlike cystic fibrosis — typically **without** characteristic *S. aureus* / *P. aeruginosa* colonization early on (PMID: 10403853).

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function mutation in SCNN1A** (frameshift/nonsense/splice/missense) → **leads to** absent or non-functional α-ENaC protein.
2. Loss of the obligatory α-subunit → **results in** failure of ENaC heterotrimer assembly/trafficking and/or absent pore conduction → **abolished amiloride-sensitive Na^+ current** at the apical membrane of sodium-absorbing epithelia. *(Demonstrated in Xenopus oocyte expression studies.)*
3. In the **distal nephron (connecting tubule/collecting duct)**: loss of apical Na^+ entry → **leads to** reduced electrogenic Na^+ reabsorption → **impaired lumen-negative transepithelial voltage → reduced driving force for K^+ and H^+ secretion**.
4. Branch → **renal salt wasting** (natriuresis, volume depletion) and **impaired K^+/H^+ excretion → hyponatremia + hyperkalemia + metabolic acidosis**.
5. Volume depletion and hyperkalemia → **stimulate renin and aldosterone → very high plasma renin activity and aldosterone**, but because the effector (ENaC) is defective → **target-organ resistance** → biochemical picture *mimics* aldosterone deficiency ("pseudo").
6. Extrarenal branch — ENaC loss in **colon** (fecal Na^+ loss), **sweat glands** (→ **high sweat Na^+/Cl^-**), **salivary glands** (→ **high saliva electrolytes**), and **airway** → multisystem phenotype.
7. Airway branch: loss of ENaC-mediated Na^+ (and fluid) absorption from airway surfaces → **excess airway surface liquid (>2× normal)** → altered mucociliary clearance and **recurrent chest congestion/cough/wheeze (CF-like)**. *(Demonstrated by in vivo airway measurements.)*
8. Severe hyperkalemia → **can cause cardiac arrhythmia/arrest** → in survivors, hypoxic-ischemic injury → **neurodevelopmental impairment**. *(Observed clinically.)*
9. Compensatory branch (partial): over time, **NCC (SLC12A3) upregulation** in the distal nephron → **partially restores Na^+ reabsorption** → clinical improvement in some patients. *(Inferred from human urinary NCC protein and mouse models.)*

Mechanistic detail

Molecular pathway: the **aldosterone–mineralocorticoid receptor (MR)–ENaC axis** (renin-angiotensin-aldosterone system, RAAS). Aldosterone normally binds MR (NR3C2), inducing transcription/activity of ENaC and Na^+/K^+-ATPase in principal cells. PHA1B1 disables the **terminal effector** (ENaC) rather than the receptor, so RAAS output is high but ineffective (PMID: 24840884).

Protein dysfunction: Cryo-EM established ENaC as a **heterotrimeric α:β:γ channel** with large extracellular "hand"-like domains (PDB 6BQN), with defined principles of assembly, proteolytic activation, and inhibition (PMID: 30251954; PMID: 32729833). Loss-of-function truncations remove pore/gating elements and abolish current; missense variants lower open probability. Truncated β-ENaC subunits can still support **partial** channel activity, explaining milder phenotypes (PMID: 20064610).

Cellular processes: defective transepithelial **ion transport** (sodium reabsorption, potassium/proton secretion) in **epithelial principal cells**; secondary **fluid homeostasis** dysregulation at airway surfaces.

Biochemical abnormality: an **ion-channel defect** (amiloride-sensitive Na^+ channel) — the defining lesion.

Suggested GO terms: sodium ion transmembrane transport (GO:0035725); regulation of sodium ion transport (GO:0002028); sodium channel activity (GO:0005272); cellular response to aldosterone (GO:1904045); potassium ion homeostasis (GO:0055075); renal sodium ion absorption (GO:0070294). **Suggested CL terms:** kidney collecting duct principal cell (CL:1000714); kidney connecting tubule epithelial cell (CL:1000768); eccrine sweat gland cell; respiratory multiciliated epithelial cell (CL:0002145); colonic epithelial cell (CL:0011108). **Suggested CHEBI terms:** sodium(1+) (CHEBI:29101); potassium(1+) (CHEBI:29103); aldosterone (CHEBI:27584); amiloride (CHEBI:2639).

Molecular profiling: No large transcriptomic/proteomic/metabolomic datasets exist for human PHA1B1; mechanistic evidence comes from targeted electrophysiology (oocyte expression) and mouse knockouts.

7. Anatomical Structures Affected

Organ level (primary): - **Kidney** (distal nephron: connecting tubule, cortical collecting duct) — UBERON:0002113; primary site of salt wasting. - **Colon / distal large intestine** — UBERON:0001155; fecal sodium loss. - **Sweat glands (eccrine)** — UBERON:0001820; elevated sweat electrolytes. - **Salivary glands** — UBERON:0001044; elevated saliva electrolytes. - **Lung / airway epithelium** — UBERON:0002048; excess airway surface liquid, CF-like disease.

Secondary/organ systems: endocrine (RAAS activation), cardiovascular (hyperkalemic arrhythmia), nervous system (post-arrest injury), skin (salt-related dermatitis/eczema).

Body systems: renal/urinary, gastrointestinal, integumentary, respiratory, cardiovascular, endocrine.

Tissue and cell level: epithelial tissue throughout; specifically Na^+-absorbing **principal cells** of the collecting duct, surface colonocytes, eccrine gland ductal cells, and **airway multiciliated epithelial cells** (ENaC localizes to cilia and regulates surface-liquid volume) (PMID: 26772908). ENaC (with CFTR) is expressed in human **epidermis and epidermal appendages including eccrine sweat glands** (PMID: 28130590).

Subcellular level: apical plasma membrane of epithelial cells (GO:0016324, apical plasma membrane); the channel traffics through ER/Golgi and is regulated at the membrane by proteases and Nedd4L-mediated ubiquitination.

Localization / lateralization: bilateral / systemic — all paired and unpaired target organs are affected symmetrically; this is not a focal or lateralized disorder.

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8. Temporal Development

Onset: Congenital/neonatal. Essentially all systemic (SCNN1A) cases present within the **first days to two weeks of life**.

> PMID: 33690157: "All children were presented within the first 2 weeks of life with life-threatening, severe hyperkalaemia and hyponatraemia. All parents were consanguineous and of South Asian, Middle Eastern or African ethnic origin."

Onset pattern: acute presentation (salt-wasting crisis / adrenal-crisis-like picture: dehydration, failure to thrive, shock) superimposed on a congenital defect.

Progression: - **Course:** chronic, **lifelong** for severe (truncating) systemic disease, punctuated by **episodic crises** triggered by illness, heat, or salt restriction. - **Rate:** variable; crises are acute, baseline disease is persistent. - **Milder variants:** some hypomorphic SCNN1A/SCNN1B cases show **transient** disease that improves/remits with age, partly via NCC compensation and reduced dependence on ENaC for salt homeostasis with growth (PMID: 23426840; PMID: 37134141).

Remission patterns: No spontaneous cure for severe disease; **treatment-controlled** stability is achievable. Partial, age-related improvement occurs with hypomorphic alleles.

Critical period: The **neonatal period** is the window of maximal vulnerability (hyperkalemic cardiac arrest) and the critical window for diagnosis and electrolyte stabilization to prevent death and neurological injury.

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9. Inheritance and Population

Inheritance: Autosomal recessive. Both *SCNN1A* alleles must be non-functional (homozygous or compound heterozygous).

Epidemiology: Very rare; systemic PHA1 prevalence is not precisely established (orders of magnitude rarer than the autosomal dominant renal form). Overall PHA1 is estimated at roughly **1 in 80,000 newborns** (all forms), with the recessive systemic form being a minority.

Penetrance: Complete for biallelic LoF genotypes (all homozygotes manifest). **Expressivity: Variable**, driven chiefly by **residual channel activity** (genotype–phenotype correlation).

> PMID: 31301676: "Patients presenting with missense mutations of SCNN1A and SCNN1B had a less marked rise in serum aldosterone suggesting preservation in sodium epithelial channel function."

Genetic anticipation: Not applicable (not a repeat-expansion disorder). **Germline mosaicism:** Not a described feature. **Consanguinity: Major contributing factor** — nearly all recessive systemic cases arise in consanguineous unions. **Founder effects:** Documented for specific ENaC alleles (e.g., shared SCNN1G splice haplotype in Indian-subcontinent families; clustered

SCNN1A variants in defined geographic populations such as a Sicilian town) (PMID: 8640238; PMID: 34134742). **Carrier frequency:** Low; not precisely quantified for SCNN1A; heterozygous carriers are asymptomatic.

Population demographics: Reported disproportionately in populations with **high consanguinity** — South Asian, Middle Eastern, North/Sub-Saharan African, and certain Mediterranean communities (PMID: 33690157). **Sex ratio:** ~1:1 (autosomal recessive; no sex predilection). **Age distribution:** Diagnosed in neonates; affected individuals survive into childhood/adulthood with treatment.

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10. Diagnostics

Laboratory tests (core): - Serum electrolytes: **hyponatremia, hyperkalemia** (LOINC 2951-2 sodium; 2823-3 potassium). - **Metabolic acidosis** (low bicarbonate; LOINC 1963-8). - **Plasma renin activity — markedly elevated** (LOINC 2915-7). - **Plasma/serum aldosterone — markedly elevated**; elevated **aldosterone:renin ratio** (LOINC 1763-2). - Urinary electrolytes: **high urine sodium**, inappropriately low urine K^+ excretion for the degree of hyperkalemia. - **Normal cortisol and normal 17-hydroxyprogesterone** — essential to exclude salt-wasting congenital adrenal hyperplasia (21-hydroxylase deficiency).

Extrarenal / differentiating tests: - **Sweat chloride test** — elevated in systemic PHA1, overlapping with cystic fibrosis. - **Saliva electrolytes** — elevated.

> PMID: 7965429: "Four patients with severe pseudohypoaldosteronism caused by multiple end-organ resistance to aldosterone had frequently recurring lower respiratory tract infections and persistently elevated sweat and saliva electrolyte values."

Genetic testing (confirmatory): - **Single-gene / gene-panel sequencing** of *SCNN1A, SCNN1B, SCNN1G* (systemic form) and *NR3C2* (renal form) is the definitive approach. - **Whole-exome sequencing (WES)** is widely used and has identified novel SCNN1A variants; **whole-genome sequencing (WGS)** or deletion/duplication analysis (MLPA/CMA) is needed to detect **gross deletions** missed by standard sequencing (PMID: 41275343). - Functional confirmation (Xenopus oocyte amiloride-sensitive current) resolves VUS, especially missense variants.

Clinical diagnostic criteria: Neonatal salt-wasting (hyponatremia + hyperkalemia + metabolic acidosis) **with high renin and aldosterone** and **poor/absent response to mineralocorticoid (fludrocortisone)**, after excluding adrenal insufficiency/CAH, plus confirmatory genetics.

Differential diagnosis: - **Salt-wasting congenital adrenal hyperplasia (21-OH deficiency)** — distinguished by elevated 17-OHP, low aldosterone, response to steroid. - **Primary adrenal insufficiency / aldosterone synthase deficiency** — low aldosterone (vs high in PHA). - **Autosomal dominant renal PHA1 (NR3C2)** — milder, kidney-restricted, normal sweat electrolytes, often improves with age (PMID: 26817011). - **Secondary (transient) PHA** — e.g., urinary tract infection/obstructive uropathy in infants. - **Cystic fibrosis** — overlapping high sweat chloride and respiratory disease; PHA1 may be clinically indistinguishable until hormonal/genetic testing.

> PMID: 22208790: "Pseudohypoaldosteronism type 1 may be indistinguishable from cystic fibrosis (CF) unless serum aldosterone, plasma renin activity, and urinary electrolytes are measured and mutation analysis rules out CF."

Notably, PHA1 and CF can rarely **coexist** (PMID: 38914525).

Screening: No routine newborn screen for PHA1B1; **cascade/carrier testing** and **prenatal/preimplantation diagnosis** are available for families with a known SCNN1A variant.

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11. Outcome / Prognosis

Mortality: Significant in severe systemic disease, driven by **hyperkalemic cardiac arrest** in the neonatal/infant period.

> PMID: 33690157: "Three children died (25%) and two (16%) had severe neurological impairment post-cardiac arrest secondary to hyperkalaemia."

Morbidity / disability: Neurological impairment in survivors of cardiac arrest (~16% in the cited cohort); chronic burden of electrolyte management; CF-like respiratory morbidity in systemic cases.

Life expectancy: With **intensive lifelong electrolyte management**, survival with **normal neurodevelopment is achievable**. Growth is often normal with adequate salt replacement (PMID: 8865238).

Disease course / complications: Recurrent dehydration/crises, failure to thrive, recurrent respiratory infections, and (in a minority) cardiac/neurological catastrophe. Systemic (SCNN1A) disease has a **poorer prognosis than renal (NR3C2) disease** and generally does **not** remit with age when the variant is a true null.

Prognostic factors: - **Genotype / residual channel activity** — the dominant prognostic determinant (truncating = severe/persistent; missense/hypomorphic = milder/transient). - **Severity of initial hyperkalemia** and timeliness of neonatal stabilization (predicts arrest/neurological outcome).

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12. Treatment

There is **no curative or disease-modifying therapy**. Management is **lifelong, supportive, and electrolyte-directed**.

Pharmacotherapy / supportive care (NCIT suggestions in parentheses): - **Oral sodium chloride supplementation**, high-dose (often up to ~10–45 mmol/kg/day), the cornerstone of therapy (NCIT:C692 Sodium Chloride). - **Sodium bicarbonate** to correct metabolic acidosis (NCIT:C29141 Sodium Bicarbonate). - **Potassium-lowering measures:** cation-exchange resins — **sodium polystyrene sulfonate (Kayexalate)** — and newer potassium binders (patiromer, sodium zirconium cyclosilicate) (NCIT:C47704 Polystyrene Sulfonate). Dietary potassium restriction. - **Acute hyperkalemia management:** insulin/glucose, beta-agonists, calcium gluconate (cardioprotection), and dialysis in refractory cases. - **Mineralocorticoids (fludrocortisone) are INEFFECTIVE** — the defect is downstream of the receptor — and do **not** form part of definitive therapy (useful diagnostically by their lack of effect). - **Indomethacin** has been used empirically in some historical cases to reduce salt loss, with reported benefit, but is not standard (PMID: 6395746).

Respiratory care: For systemic patients with CF-like airway disease, **CF-style airway clearance and infection management** may be required.

> PMID: 7965429: "Patients with pseudohypoaldosteronism may require treatment similar to that for cystic fibrosis to prevent long-term respiratory complications."

Advanced/experimental therapeutics: No approved gene, cell, RNA-based, or targeted therapy exists. Proof-of-concept laboratory work shows **pharmacological rescue of channel function** — e.g., the **TNF lectin-like domain (AP301/solnatide)** restored ENaC function in PHA1B frameshift mutants in vitro (PMID: 28611771) — but this is not a clinical therapy for PHA1B1.

Treatment strategy: Individualized titration of sodium supplementation, acidosis correction, and potassium control to normalize electrolytes and support growth; multidisciplinary follow-up (endocrinology, nephrology, pulmonology, genetics). Personalized intensity is guided by **genotype/residual activity** — milder variants may need less and may be weaned as the child grows.

Treatment outcomes: With adherence, electrolytes and renin/aldosterone can normalize and growth proceeds normally; however, severe cases demand large daily supplement doses and remain at risk of crises.

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13. Prevention

The disease itself cannot be prevented (congenital genetic cause); prevention focuses on **avoiding complications** and **reproductive risk reduction**.

- **Primary prevention:** Not applicable for disease occurrence. **Genetic counseling** for consanguineous couples and families with a known variant; **carrier testing**, **prenatal diagnosis**, and **preimplantation genetic diagnosis** allow informed reproductive choices and early preparedness (PMID: 34134742).
- **Secondary prevention:** Early recognition of neonatal salt-wasting and prompt genetic confirmation to start therapy before a catastrophic hyperkalemic event; cascade screening of at-risk siblings.
- **Tertiary prevention:** Rigorous electrolyte management, sick-day rules (increased salt, early medical review during illness/heat), and respiratory surveillance to prevent crises, cardiac arrest, neurological injury, and chronic lung disease.

- **Immunization / public health / environmental interventions:** Not applicable to the genetic cause; routine childhood immunization and respiratory-infection precautions are prudent given airway involvement.

14. Other Species / Natural Disease

- **Taxonomy:** Human disease (*Homo sapiens*, NCBI Taxon 9606). Naturally occurring PHA1B1 is not a recognized spontaneous disease in companion animals or wildlife (no OMIA entry for a natural SCNN1A-PHA phenotype of veterinary importance).
- **Orthologous genes:** *Scnn1a* in mouse (NCBI Gene 20276, chr 6) and rat (*Scnn1a*, NCBI Gene 25122). ENaC subunit genes are **deeply conserved** across vertebrates — orthologs appear from jawless fish onward, and most tetrapods carry four ENaC paralogs (PMID: 26772908).
- **Comparative biology:** The aldosterone–ENaC axis is evolutionarily conserved, making rodent models highly informative. Complete α-ENaC knockout in mice is **neonatally lethal from defective lung fluid clearance**, underscoring the channel's conserved role in airway as well as renal sodium/fluid handling.
- **Transmission:** None — non-infectious, non-zoonotic genetic disorder.

15. Model Organisms

Mouse (Mus musculus) — the principal model. - **Constitutive α-ENaC (*Scnn1a*) knockout:** dies within ~40 h of birth from failure to clear lung fluid — demonstrating ENaC's essential role in perinatal airway fluid absorption and modeling the respiratory dimension of systemic PHA1. - **Inducible, nephron-specific α-ENaC knockout (Scnn1a^Pax8/LC1):** recapitulates renal PHA-1 in adults — **hyperkalemia and body-weight loss on a regular diet**, rescued by a high-Na^+/low-K^+ diet.

> PMID: 26701978: "We generated adult inducible nephron-specific αENaC-knockout mice (Scnn1a(Pax8/LC1)) that exhibit hyperkalemia and body weight loss when kept on a regular-salt diet, thus mimicking PHA-1."

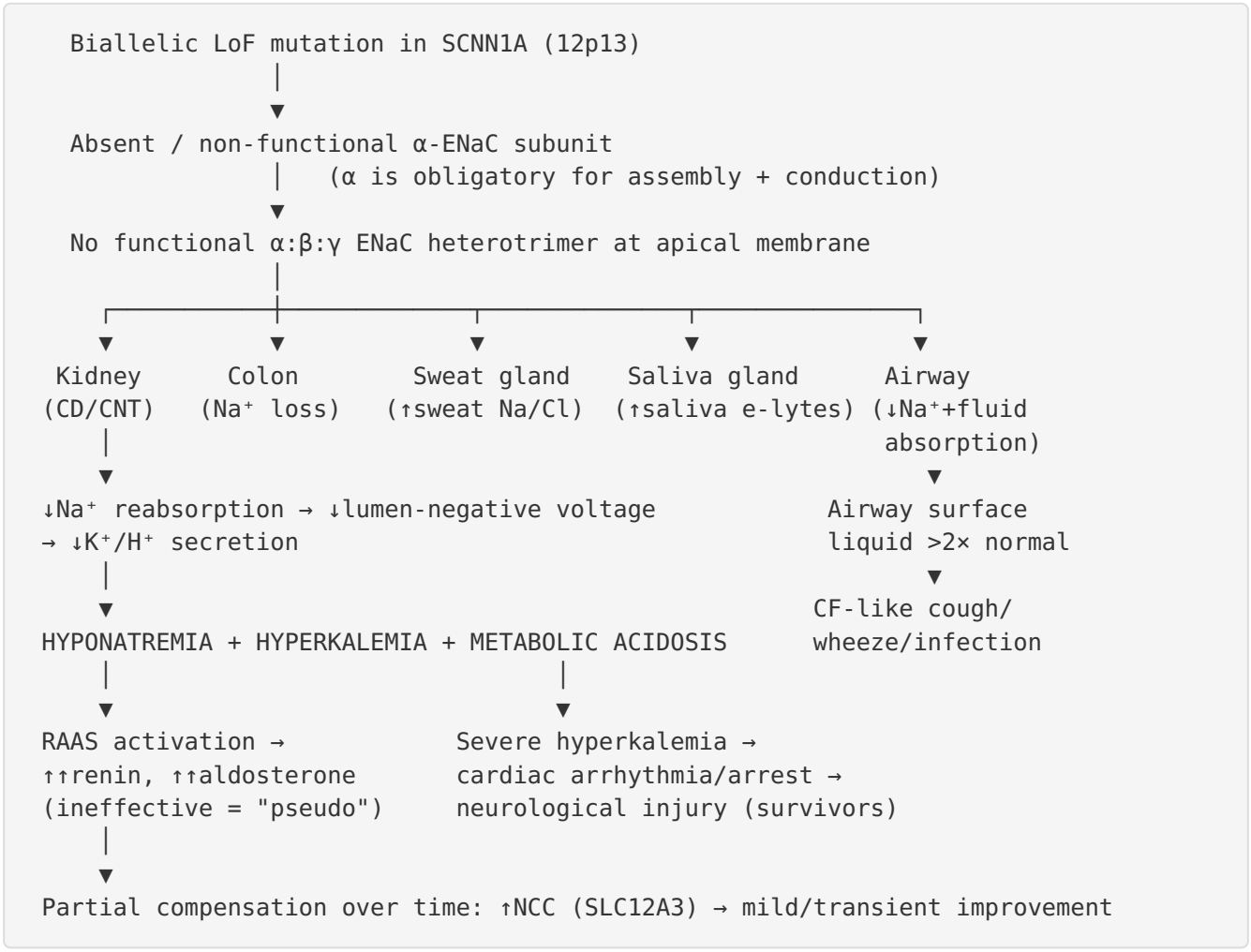
- Importantly, these mice reveal a **compensatory mechanism**: on a regular diet NCC is paradoxically downregulated despite high aldosterone, and a rescue diet normalizes electrolytes and restores NCC — dovetailing with the human finding of NCC-mediated compensation (PMID: 20376516).
- **Conditional (floxed) alleles for Scnn1b and Scnn1g** exist to dissect β- and γ-subunit roles, since their constitutive knockouts are also perinatally lethal (PMID: 19036848).

In vitro / cellular models: - **Xenopus laevis oocyte expression** of mutant ENaC subunits is the standard functional assay for pathogenicity and residual activity (e.g., truncated β-ENaC retains partial activity; γ-A635V reduces current ~30%) (PMID: 20064610; PMID: 40153144).

Model characteristics: Inducible nephron-specific knockouts **faithfully reproduce the renal electrolyte phenotype** and the diet-dependent compensation; constitutive knockouts capture the **lethal pulmonary** phenotype. **Limitations:** no single mouse fully reproduces the complete human multisystem spectrum simultaneously (renal + sweat/saliva + chronic survivable lung disease), and perinatal lethality of global knockouts necessitates conditional/inducible strategies.

Resources: MGI (*Scnn1a*, *Scnn1b*, *Scnn1g*), IMPC/KOMP conditional alleles, IMSR.

Mechanistic Model (Synthesis)



Upstream vs downstream: The mutation and loss of ENaC conduction are **upstream**; renal/colonic salt wasting and the electrolyte triad are **proximal downstream** effects; RAAS activation, crises, cardiac/neurological events, and respiratory disease are **distal downstream** consequences. NCC upregulation is a **compensatory feedback branch**.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
8589714	ENaC subunit mutations cause PHA1	Human genetics (landmark)	Causal LoF in α/β ENaC; recessive inheritance
8640238	Homozygosity mapping of PHA1	Human genetics (landmark)	Loci at 12p (SCNN1A) & 16p; founder haplotype
38963175	Novel SCNN1A mutation, PHA1B	Human case / review	PHA1B from ENaC subunits; ~45 variants
41457049	PHA1B2 twins (SCNN1B)	Human case	Gene triad; severity; SCNN1A most common
33690157	AR-PHA1 clinical cohort	Human cohort (12 pts, 37 y)	Neonatal onset, consanguinity, 25% mortality
31301676	Genotype–phenotype in PHA1	Human cohort	Missense → residual activity, milder disease
10403853	Airway ENaC dysfunction in PHA1	Human physiology	Abolished airway Na^+ absorption; 2× ASL
22208790	Pulmonary complications review	Review	PHA1 vs CF differentiation
7965429	Sweat/saliva electrolytes in PHA	Human case series	Elevated sweat/saliva; CF-like respiratory care
28130590	ENaC/CFTR in epidermis	Human expression	Molecular basis of sweat phenotype
20376516	NCC upregulation in AR-PHA1	Human	Compensatory NCC mechanism
26701978	Nephron-specific αENaC KO mouse	Mouse model	Recapitulates renal PHA-1
19036848	Conditional Scnn1b/ Scnn1g	Mouse model	Tools for subunit dissection
20064610	Truncated β-ENaC partial activity	In vitro (oocyte)	Residual-activity genotype–phenotype
40153144	γ-ENaC A635V function	In vitro + human	Missense reduces current ~30%

PMID	Title (abbrev.)	Evidence type	Supports
30251954	Cryo-EM ENaC structure	Structural	Heterotrimeric architecture
32729833	ENaC assembly/activation	Structural	Assembly, proteolytic activation
26772908	ENaC family review	Review	Tissue distribution, phylogeny, disease
24840884	Aldosterone function disorders	Review	RAAS–ENaC axis, PHA genetics
28611771	TNF lectin-like domain rescue	In vitro	Experimental channel rescue
41275343	Gross deletion in SCNN1A	Human case	Deletion-detection diagnostic gap
34134742	SCNN1A splice mutation, Sicily	Human case	Founder clustering; prevention via NGS
37134141	Mild/transient AR-PHA1	Human cohort	Milder phenotype spectrum
23426840	SCNN1B splice mutation, Turkey	Human case	Age-related improvement mechanism
8865238	Severe PHA twins	Human case	Lifelong management; normal growth achievable
6395746	PHA review / indomethacin	Human case	Historical salt/indomethacin therapy
26817011	Renal PHA1 (NR3C2)	Human case	Distinguishes renal vs systemic PHA1
38914525	PHA1 coexisting with CF	Human case	PHA1/CF overlap and coexistence

Limitations and Knowledge Gaps

1. **SCNN1A-specific vs pooled data.** Much of the clinical and mechanistic literature pools the three ENaC subunit genes under "PHA1B." Pure SCNN1A-only cohorts are small, and some cited cohorts include SCNN1G/SCNN1B patients.

2. **No large-scale omics.** No transcriptomic, proteomic, metabolomic, or single-cell datasets exist specifically for human PHA1B1; mechanism rests on targeted electrophysiology and mouse genetics.
3. **Epidemiology imprecise.** True prevalence/incidence and carrier frequency of SCNN1A-PHA1B1 are not well quantified; estimates are extrapolated from all-PHA1 figures.
4. **Structure–function for specific variants.** Only a minority of reported SCNN1A variants have functional characterization; many remain inferred LoF.
5. **No disease-modifying therapy.** Promising in vitro channel-rescue approaches (e.g., solnatide/AP301) have not been translated to PHA1B1 clinical trials.
6. **Quality-of-life data absent.** No validated QoL instrument data for this rare disorder.
7. **Deletion detection gap.** Standard sequencing can miss gross SCNN1A deletions, risking under-diagnosis.

Proposed Follow-up Experiments / Actions

1. **Assemble an SCNN1A-only genotype–phenotype registry** correlating variant class (null vs hypomorphic) and functional residual activity with mortality, salt-requirement, and respiratory outcomes.
2. **Systematic functional assays** (oocyte/mammalian electrophysiology + cryo-EM-guided modeling) for all reported SCNN1A VUS to firm up ACMG classification and residual-activity predictions.
3. **Ensure deletion/duplication analysis (MLPA/CMA/WGS)** in genetically unconfirmed clinically typical cases to close the gross-deletion diagnostic gap.
4. **Translate channel-rescue pharmacology** (e.g., solnatide/AP301, small-molecule ENaC potentiators) toward preclinical PHA1B1 models and, if promising, early-phase trials.
5. **Characterize the extrarenal burden** (longitudinal sweat/saliva electrolytes, pulmonary function, airway microbiome) to define when CF-style respiratory care improves outcomes.
6. **Investigate NCC-targeted or dietary strategies** that exploit the compensatory NCC branch to reduce salt-supplement burden.
7. **Develop a humanized/hypomorphic knock-in mouse** reproducing the survivable multisystem phenotype for therapeutic testing.

Report compiled from 7 confirmed findings and 36 reviewed papers across a 5-iteration autonomous investigation. Evidence source types are indicated throughout (human clinical, model organism, in vitro, structural/computational).
